Modeling Customer Churn

A Machine Learning approach to proactive retention

The Problem Statement

Every quarter, roughly one in five 'FOCUS' customers stop shipping. The challenge is identifying the hidden signals that lead to customer churn, to help shift the company from reactive loss to proactive, data-driven retention.

> **Churn is defined as a Focus customer who places no shipments in the following quarter, and not a temporary dip in volume, but a full stop.**
> This definition creates a clean Yes/No label that the model learns from, and eliminates ambiguity about what counts as at-risk.

With the model tasked to generate a ranked watchlist before the quarter closes, not after.

The Data

The model draws on [redacted] quarters of internal freight history [redacted] pulling from four source tables and compressing over [redacted] million freight bills into a single quarterly summary row per customer.

Source	Records	Metric Extracted	Role in Model
Freight Bills	4.1M	Volume, weight, miles, charges	Core shipping behavior
OSD Exceptions	6.3M	Over, short & damage incidents	Service quality signal
Late Shipments	2.1M	Days late, late pickup flag	Delivery reliability
Claims	198K	Claim amount, amendments	Dispute & settlement friction

Key Engineered Features

Raw records were transformed into seven predictive signals per customer per quarter:

- Shipping Momentum - compares piece volume in month 3 vs month 1 of the same quarter. A negative value means the customer is shipping less as the quarter progresses.
- OSD Score - average over, short, and damage incidents per freight bill.
- Charge per Mile - cost efficiency metric; a proxy for freight density and pricing sensitivity.
- Pieces per Week - absolute weekly volume, normalized by weeks active in the quarter.
- Short/Long-Haul Mix - proportion of shipments above vs below 500 miles, reflecting lane competition exposure.
- Service Delays - average days late and late pickup rate.

- Claims Activity - average claim value and amendment count per bill.

Four Customer Personas

K-Means clustering on the scaled feature set identified four naturally occurring customer profiles. These are not model inputs, they are descriptive groups that emerged from the data.

Cluster	Churn Rate	Short Haul	Momentum	Profile
Cluster 1	25.8%	8%	−0.42	[redacted]
Cluster 2	6.1%	49%	+0.06	[redacted]
Cluster 3	28.0%	95%	+0.09	[redacted]
Cluster 4	19.2%	9%	+0.85	[redacted]

> Notable: Cluster 2, the Stable Core, has nearly even short/long-haul mix, the lowest OSD scores, and flat but positive momentum. Protecting these accounts is as important as identifying those at risk.

Modeling Approach

Why Two Separate Models

High-frequency and low-frequency shippers behave differently and churn for different reasons. A single model would be forced to average across those behaviors. Two segment-specific models allow the algorithm to learn each population independently.

Segment	Accounts	Training Churn Rate	Threshold Strategy
High-Volume (≥1.04 shipments/week)	[redacted]	4.8%	75% recall floor
Low-Volume (<1.04 shipments/week)	[redacted]	41.8%	90% recall floor

Algorithm: XGBoost

XGBoost was chosen for its ability to handle structured tabular data, produce feature importance rankings, and support SHAP-based explainability. Unlike deep learning approaches, every prediction can be traced back to the specific features that drove it, making the output defensible in a business context.

Handling Class Imbalance : SMOTE

Because most customers do not churn in any given quarter, a naive model would simply predict retention for everyone and achieve misleading accuracy. SMOTE (Synthetic Minority Oversampling) was applied to the training data to generate synthetic churn examples, capped at 20% of the majority class to avoid over-correction.

Threshold Tuning

Rather than using a fixed probability cutoff, the model sweeps all thresholds and selects the one that maximizes the F1 score subject to a minimum recall constraint. This is the strategic dial; it can be adjusted based on Sales capacity and risk appetite.

Top Churn Drivers

#	Feature	Importance	What It Means
1	**Shipping Momentum**	55% (HSF) / 27% (LSF)	Volume is declining within the quarter, and the drop is happening before the quarter ends
2	**OSD Score**	14% (HSF) / 18% (LSF)	Elevated damage, shortage, or over-count incidents signal service friction that precedes switching
3	**Charge per Mile**	16% (LSF)	High cost relative to distance suggests price sensitivity, particularly acute for low-frequency shippers
4	**Pieces per Week**	10% (HSF)	Absolute volume stability, a sustained drop in weekly pieces, separate from momentum, reinforces risk
5	**Haul Mix (Short/Long)**	5–8.5%	Short-haul accounts face more carrier competition; long-haul accounts are sensitive to transit reliability

Model Results: Q4 2025 Test

Segment	AUC	Recall	Flagged
Low-Volume (90% recall floor)	0.711	90.0%	4,675
High-Volume (75% recall floor)	0.877	75.5%	1,894
Combined	-	87.9%	6,569

A note on the High-Volume model: the recall floor was intentionally set at 75% rather than 90%. The High-Volume segment has a very low base churn rate (4.8%), which means catching 90% of churners requires flagging nearly the entire account base - generating excessive false alarms on your most important accounts. The current setting catches 372 actual churners while limiting false flags to 1,522.

Using the Output

Each quarter, the model produces a prioritised CSV with one row per flagged account, including:

- Account ID and quarter scored
- Churn probability score and severity label (HIGH / MEDIUM / LOW)
- Volume segment (High-Volume / Low-Volume) and customer cluster

- Key inline metrics in real-world values (shipments/week, pieces/week, haul mix)
- Top 3 churn drivers with a specific recommended action for each

Severity	Probability	Accounts (Q4 2025)	Sample Recommended Response
HIGH	≥ 0.65	879	Immediate Sales outreach, prioritise within the week
MEDIUM	0.40 – 0.64	3,655	Scheduled review, address the root cause driver by driver
LOW	At threshold – 0.39	2,035	Monitor, flag if another negative signal emerges next quarter

Customer ID	Churn Prob.	Alert	Recommended Action
[redacted]	0.885	Momentum @ -0.42 + Short-Haul @ 92%: Volume erosion in highly competitive regional lanes.	Defend the account with a targeted regional pricing incentive.
[redacted]	0.722	Charge @ $1.52 + OSD Score @ 1.9: Premium pricing coupled with elevated service friction.	Initiate a rate review alongside a formal service recovery plan.
[redacted]	0.684	Days Late @ 3.1 + Long-Haul @ 88%: Critical service failures on high-margin freight.	Escalate to Operations for an immediate logistics performance audit.
[redacted]	0.616	Piece Momentum @ -0.58: Significant and accelerated volume diversion detected.	Immediate outreach to identify and counter shifts to secondary providers.

Customer ID	Churn Prob.	Segment	Driver 1	Driver 2
[redacted]	0.909	Low Ship Freq	Momentum drop	Exceptions threshold (OSD)
[redacted]	0.902	High Ship Freq	Exceptions threshold (OSD)	Weekly pallet count low
[redacted]	0.882	High Ship Freq	Above avg. cost_per_mile	Momentum drop
[redacted]	0.874	High Ship Freq	Momentum drop	Weekly pallet count low
[redacted]	0.761	Low Ship Freq	Exceptions threshold (OSD)	Above avg. cost_per_mile
[redacted]	0.722	Low Ship Freq	Momentum drop	Exceptions threshold (OSD)

Scope & Future Enhancements

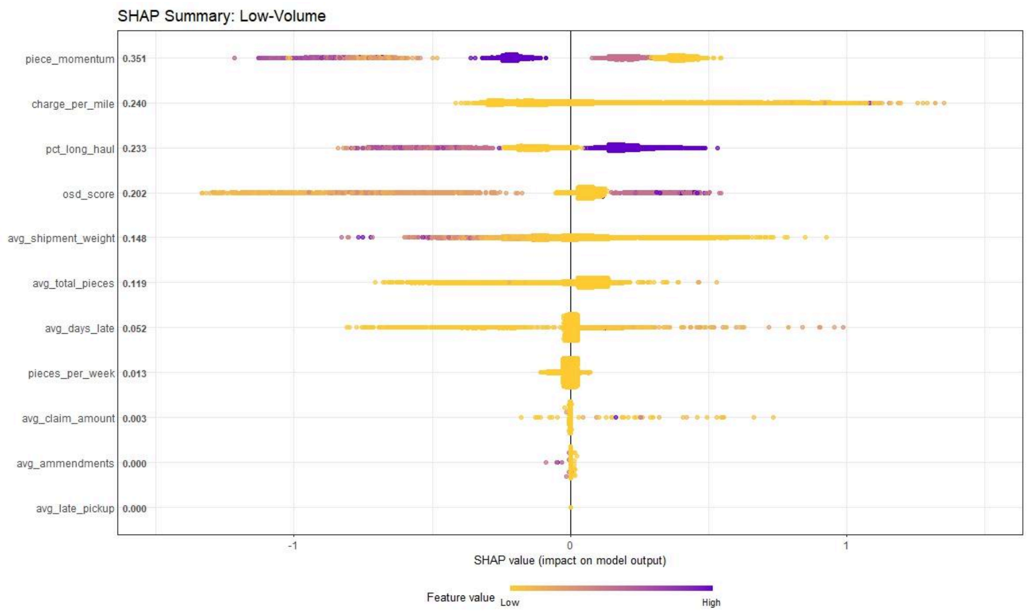
The current model treats each customer-quarter as an independent observation, using only internal freight and service data. The following would meaningfully extend its predictive power:

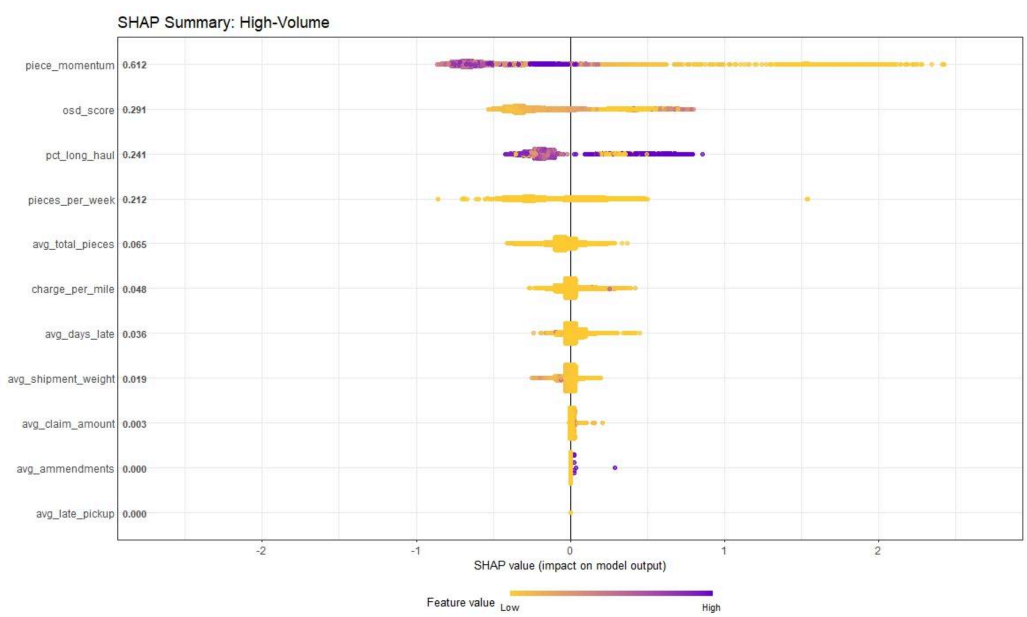
- Transit days data, when available, transit time consistency is a strong candidate for lifting the High-Volume AUC further.

- Cross-quarter trajectory features, lag variables such as prior-quarter momentum, and consecutive quarters of volume decline, once the independent-observation constraint is relaxed.
- External signals, regional economic indicators, fuel indices, or carrier capacity data could contextualise changes in shipping behavior.
- Deeper learning models, once interpretability requirements are established and the output is operationalised, ensemble or sequence models could be evaluated.

Understanding the SHAP Charts

SHAP (SHapley Additive exPlanations) values show not just which features matter, but how much each one pushed a specific prediction up or down. Each dot in the chart represents one account in the test set.





How to read the chart

- Horizontal position - a dot to the right of zero increases predicted churn probability; a dot to the left reduces it.
- Dot colour - yellow dots represent accounts with a low feature value; purple dots represent accounts with a high feature value.
- Feature order - features are ranked top to bottom by their overall contribution to the model's predictions. The number next to each feature name is its mean absolute SHAP value.

What the charts tell us

Low-Volume accounts. Momentum, cost-per-mile, and long-haul exposure share influence roughly equally; an account showing decline on multiple signals simultaneously is significantly more at risk.

High-Volume accounts. Momentum accounts for 61% of the model's explanatory power. When a high-frequency shipper starts pulling back within a quarter, the model treats it as the dominant churn signal.

Across both segments. Long-haul exposure ranks third in both models. Accounts with a high proportion of freight moving over 500 miles are more sensitive to service consistency and transit reliability.

The Team

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